POLITECHNIKA WROCŁAWSKA Wydział Informatyki i Telekomunikacji	Wydział: Informatyki i Telekomunikacji Kierunek: Cyberbezpieczeństwo Rok Akademicki: 2025/2026 Rok studiów, semestr: *2, 2* Grupa: *2*
Ochrona Systemów Operacyjnych – Laboratorium 2	
Prowadzący: mgr inż. Bartłomiej Czesak	Autor: 1. *Patryk Wesołowski*
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Data oddania sprawozdania: *24.03.2026*	

Zadanie 1.1 Instalacja Hashicore Vault i wstępna konfiguracja

```
wesoly@Ubuntu:~$ wget -O- https://apt.releases.hashicorp.com/gpg | gpg --dearmor | sudo
 tee /usr/share/keyrings/hashicorp-archive-keyring.gpg
--2026-03-18 16:55:23--  https://apt.releases.hashicorp.com/gpg
Resolving apt.releases.hashicorp.com (apt.releases.hashicorp.com)... 3.174.230.110, 3.1
74.230.38, 3.174.230.14, ...
Connecting to apt.releases.hashicorp.com (apt.releases.hashicorp.com)|3.174.230.110|:44
3... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3980 (3.9K) [binary/octet-stream]
Saving to: ‘STDOUT’

-                   100%[=======================>]   3.89K  --.-KB/s    in 0s

2026-03-18 16:55:23 (2.43 GB/s) - written to stdout [3980/3980]

c◆◆◆◆◆◆◆◆|◆jż◆m:◆z◆◆◆C#◆◆◆◆2UT\:◆~◆◆[◆◆O◆}◆Y◆*◆◆8uU(◆◆◆◆◆◆◆◆$◆◆Q◆◆qG7◆◆S`◆yη◆Y◆[jw◆◆b◆g◆◆◆Z◆
◆◆w◆?◆◆l0◆◆◆V◆F◆◆v◆{
    :f6◆◆Kx◆!◆◆◆◆!◆h◆d◆◆I◆◆
                          6Xg{4◆=◆◆YZ.◆T◆s◆◆{◆E◆◆◆◆◆◆>◆ R◆◆◆◆{◆*◆◆K◆◆◆◆◆◆◆k<◆Mq◆)◆◆c0◆
◆Tj:/ȳh◆h◆◆◆Nf◆◆◆◆◆c◆◆c◆◆◆◆W{5!◆Q◆◆◆◆6&◆Y
                                        ◆b[◆◆¥◆◆◆◆
                                                  ◆R◆◆◆◆◆◆◆◆◆a◆}k57◆◆◆R]◆◆ ◆◆◆◆◆#◆z◆◆◆◆~
'd
```

Rysunek 1 Pobranie kluczy

```
wesoly@Ubuntu:~$
echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.rel
eases.hashicorp.com $(lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/hashico
rp.list
deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.
hashicorp.com noble main
```

Rysunek 2 Dodanie repozytorium

```
wesoly@Ubuntu:~$ sudo apt update && sudo apt install vault

Get:1 https://apt.releases.hashicorp.com noble InRelease [12.9 kB]
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Get:4 https://apt.releases.hashicorp.com noble/main amd64 Packages [225 kB]
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 238 kB in 1s (434 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
187 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following NEW packages will be installed:
  vault
0 upgraded, 1 newly installed, 0 to remove and 187 not upgraded.
Need to get 171 MB of archives.
After this operation, 517 MB of additional disk space will be used.
Get:1 https://apt.releases.hashicorp.com noble/main amd64 vault amd64 1.21.4-1 [171
MB]
Fetched 171 MB in 15s (11.5 MB/s)
Selecting previously unselected package vault.
(Reading database ... 196552 files and directories currently installed.)
Preparing to unpack .../vault_1.21.4-1_amd64.deb ...
Unpacking vault (1.21.4-1) ...
Setting up vault (1.21.4-1) ...
Generating Vault TLS key and self-signed certificate...
...+...+...+.....+...+......+....+..+.......+..+....+..+...+.......+................
.+......+...........+.......+..+.+........+....+...+.....+.......+...+..+.......+..
.+....+......+........+........+...........+........+..+...+.+.......+...+...
```

VMware Tools

Rysunek 3 Instalowanie Hashicorp vault

```
wesoly@Ubuntu:~$ vault -version
Vault v1.21.4 (ffe7023c481dc1ea2d8550bbaca8d85f8e611e0b), built 2026-03-04T17:40:05Z
```

Rysunek 4 Sprawdzenie poprawności instalacji

```
wesoly@Ubuntu:~$ vault server -dev -dev-root-token-id=root
==> Vault server configuration:

Administrative Namespace:
             Api Address: http://127.0.0.1:8200
                     Cgo: disabled
         Cluster Address: https://127.0.0.1:8201
   Environment Variables: CLUTTER_DISABLE_MIPMAPPED_TEXT, COLORTERM, DBUS_SESSION_BUS_ADDRESS, DEBUGINFOD_URLS, DESK
ETUP_DISPLAY, GNOME_SHELL_SESSION_MODE, GNOME_TERMINAL_SCREEN, GNOME_TERMINAL_SERVICE, GOTRACEBACK, GSM_SKIP_SSH_AGE
 LESSOPEN, LOGNAME, LS_COLORS, MEMORY_PRESSURE_WATCH, MEMORY_PRESSURE_WRITE, PATH, PWD, QT_ACCESSIBILITY, QT_IM_MODU
 TERM, USER, USERNAME, VTE_VERSION, WAYLAND_DISPLAY, XAUTHORITY, XDG_CONFIG_DIRS, XDG_CURRENT_DESKTOP, XDG_DATA_DIRS
DESKTOP, XDG_SESSION_TYPE, XMODIFIERS, _
              Go Version: go1.25.7
              Listener 1: tcp (addr: "127.0.0.1:8200", cluster address: "127.0.0.1:8201", disable_request_limiter: "
", max_json_object_entry_count: "10000", max_json_string_value_length: "1048576", max_request_duration: "1m30s", max
               Log Level:
                   Mlock: supported: true, enabled: false
           Recovery Mode: false
                 Storage: inmem
                 Version: Vault v1.21.4, built 2026-03-04T17:40:05Z
             Version Sha: ffe7023c481dc1ea2d8550bbaca8d85f8e611e0b

==> Vault server started! Log data will stream in below:

2026-03-18T17:17:47.722+0100 [INFO]  proxy environment: http_proxy="" https_proxy="" no_proxy=""
2026-03-18T17:17:47.722+0100 [INFO]  incrementing seal generation: generation=1
2026-03-18T17:17:47.722+0100 [WARN]  no `api_addr` value specified in config or in VAULT_API_ADDR; falling back to d
2026-03-18T17:17:47.723+0100 [INFO]  core: Initializing version history cache for core
2026-03-18T17:17:47.723+0100 [INFO]  events: Starting event system
2026-03-18T17:17:47.737+0100 [INFO]  core: security barrier not initialized
2026-03-18T17:17:47.737+0100 [INFO]  core: security barrier initialized: stored=1 shares=1 threshold=1
2026-03-18T17:17:47.737+0100 [INFO]  core: post-unseal setup starting
2026-03-18T17:17:47.746+0100 [INFO]  core: loaded wrapping token key
2026-03-18T17:17:47.746+0100 [INFO]  core: successfully setup plugin runtime catalog
2026-03-18T17:17:47.746+0100 [INFO]  core: successfully setup plugin catalog: plugin-directory=""
2026-03-18T17:17:47.746+0100 [INFO]  core: no mounts; adding default mount table
```

Rysunek 5 Uruchomienie serwera

```
wesoly@Ubuntu:~$ export VAULT_ADDR='http://127.0.0.1:8200'
wesoly@Ubuntu:~$
```

Rysunek 6 Ustawienie zmiennej

```
wesoly@Ubuntu:~$ cat pki-policy.hcl
path "sys/mounts/*" {
  capabilities = [ "create", "read", "update", "delete", "list" ]
}


path "sys/mounts" {
  capabilities = [ "read", "list" ]
}

path "pki*" {
  capabilities = [ "create", "read", "update", "delete", "list", "sudo", "patch" ]
}

wesoly@Ubuntu:~$ vault policy write pki_policy pki-policy.hcl
Success! Uploaded policy: pki_policy
```

Rysunek 7 Dodanie polityki

Zadanie 1.2 Generowanie lokalnego CA

```
wesoly@Ubuntu:~$ vault secrets enable pki
Success! Enabled the pki secrets engine at: pki/
```

Rysunek 8 Włączenie pki

```
wesoly@Ubuntu:~$ vault secrets tune -max-lease-ttl=87600h pki
Success! Tuned the secrets engine at: pki/
```

Rysunek 9 Ustawienie maksymalnego czasu ważności certyfikatów

```
wesoly@Ubuntu:~$ vault write -field=certificate pki/root/generate/internal common_name="example.c
om" issuer_name="root-2026" key_bits=4096 signature_bits=256 ttl=87600h > root_2026_ca.crt
```

Rysunek 10 Utworzenie CA

```
wesoly@Ubuntu:~$ vault write pki/config/urls \
> issuing_certificates="$VAULT_ADDR/v1/pki/ca" \
> crl_distribution_points="$VAULT_ADDR/v1/pki/crl"
Key                              Value
---                              -----
crl_distribution_points          [http://127.0.0.1:8200/v1/pki/crl]
delta_crl_distribution_points    []
enable_templating                false
issuing_certificates             [http://127.0.0.1:8200/v1/pki/ca]
ocsp_servers                     []
wesoly@Ubuntu:~$
```

Rysunek 11 Konfiguracja URL dla CA i CRL

```
wesoly@Ubuntu:~$ vault write pki/roles/user-cert allow_any_name=true allow_bare_domains=true enforce_hostnames=false allow_subdomain
s=true max_ttl="8760h" key_bits=4096 signature_bits=256 hash_algorithm="sha2-256" key_usage="DigitalSignature,KeyEncipherment" ext_k
ey_usage="ClientAuth,EmailProtection"
WARNING! The following warnings were returned from Vault:

  * Endpoint ignored these unrecognized parameters: [hash_algorithm]

Key                                 Value
---                                 -----
allow_any_name                      true
allow_bare_domains                  true
allow_glob_domains                  false
allow_ip_sans                       true
allow_localhost                     true
```

Rysunek 12 Stworzenie roli odpowiedzialnej za podpisywanie certyfikatów

Zadanie 2 Generowanie wniosku

```
wesoly@Ubuntu:~$ openssl req -new -newkey rsa:4096 -nodes \
> -keyout user.key \
> -out user.csr \
> -sha256 \
> -subj "/CN= Patryk Wesolowski/emailAddress=283879@student.pwr.edu.pl"
....+........+......+.......+...+..+........+.+...+......+........+..+....+...+...+..
+......+...................+..+..........+........+.++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++*..+....+..........+++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++*.......+..........+...+..................+...+........+.
```

Rysunek 13 CSR

Zadanie 3 Podpisanie certyfikatu

```
wesoly@Ubuntu:~$ vault write pki/sign/user-cert csr=@user.csr format=pem_bundle ttl=8760h
Key                 Value
---                 -----
authority_key_id    f8:86:a0:6e:28:06:7b:2f:27:8d:4d:db:ba:26:d2:de:e4:7c:10:92
```

Rysunek 14 Podpisanie certyfikatu

Zadanie 4 Weryfikacja certyfikatu

```
wesoly@Ubuntu:~$ openssl verify -CAfile ca.pem cert.pem
cert.pem: OK
```

Rysunek 15 Weryfikacja certyfikatu

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            04:66:96:48:83:47:f8:af:b6:cc:71:c3:dc:1b:00:fb:1d:61:82:ce
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: CN = example.com
        Validity
            Not Before: Mar 18 17:26:53 2026 GMT
            Not After : Mar 18 17:27:23 2027 GMT
        Subject: CN = " Patryk Wesolowski"
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (4096 bit)
                Modulus:
                    00:95:27:3f:2d:c0:c1:83:b6:63:ac:a6:bb:b4:43:
                    f9:11:9f:1d:20:45:c6:af:2e:1e:27:57:ad:41:53:
                    c2:d7:0c:87:28:c6:16:cb:4f:8a:2c:43:64:5f:3a:
                    7f:91:a6:be:80:08:39:76:eb:e0:13:83:0c:08:38:
                    e6:f3:3e:5a:3e:f9:a6:dd:c5:2b:9d:a0:08:68:55:
                    4b:81:50:86:17:98:8e:23:47:7a:cc:e4:15:db:80:
                    26:14:c7:76:7c:77:c2:d1:e2:45:87:39:56:aa:39:
                    6f:5f:aa:60:a3:71:f5:ab:22:b7:19:33:22:ae:4e:
                    9b:21:fc:5b:f0:a4:f9:4d:a4:9b:af:c1:c9:3e:54:
                    e4:99:51:56:a7:72:18:c8:4d:88:af:cb:b7:b3:d0:
                    20:e1:ca:36:94:f2:a6:f5:1e:7e:4b:52:e5:32:98:
                    aa:7a:74:75:a7:7d:1a:9c:ee:cb:26:e4:a6:2f:e7:
                    ec:3f:ec:67:5e:8f:b0:b4:38:db:de:17:9d:1c:1a:
                    61:1c:b2:42:a3:d3:10:fb:f5:81:f5:06:4f:50:26:
                    38:3a:7a:76:9e:7c:46:b8:20:1e:76:1a:a6:42:b2:
```

```
            45:98:98:ff:7c:b0:41:94:c2:6b:38:d9:52:97:98:
            8a:77:64:d8:bf:c1:21:5a:bb:ef:76:4e:46:0a:e9:
            6e:50:bd:ff:f0:d2:1f:d5:22:ed:7c:e9:88:aa:85:
            19:d8:81:de:eb:0a:f0:d4:b4:39:c9:67:93:e3:e3:
            03:0f:57:f1:b9:3c:1e:59:42:e6:ac:71:19:de:d1:
            b4:29:12:e7:e7:5e:f9:31:9e:6e:84:c6:15:e1:88:
            85:56:c8:d8:54:ca:57:03:e2:ff:b2:cf:87:d0:7f:
            3d:ea:93:cc:bd:cf:85:28:aa:41:63:66:7f:de:2b:
            af:91:32:c0:da:ae:3b:83:e6:4b:c8:b1:dd:ba:d9:
            98:1a:66:91:b1:40:af:fe:8c:c8:1b:e2:8d:11:d7:
            16:fd:bc:d5:2e:2a:bb:a0:7a:71:75:da:f2:40:23:
            6d:70:89:bf:fa:21:a8:d5:7a:55:2b:d6:1d:29:b5:
            51:58:a1:d8:3b:32:53:cb:fb:6c:96:07:40:01:a6:
            81:fc:ba:6e:a6:14:19:c4:8b:79:29:22:81:29:38:
            d5:80:66:ca:3e:33:bb:27:89:22:cb:ee:8a:79:ad:
            33:bd:59:bb:c2:5d:57:59:2e:c7:5e:13:bd:be:71:
            b3:01:d0:66:9f:4a:80:96:cc:c5:de:3f:b3:74:85:
            c8:15:db:2f:11:9d:8c:14:74:93:e3:aa:eb:49:88:
            8e:f2:d9:49:bb:84:9b:fa:21:a2:12:03:15:d0:a0:
            cb:8e:2d
        Exponent: 65537 (0x10001)
    X509v3 extensions:
        X509v3 Key Usage: critical
            Digital Signature, Key Encipherment
        X509v3 Extended Key Usage:
            TLS Web Server Authentication, TLS Web Client Authentication, E-mail Protection
        X509v3 Subject Key Identifier:
            E5:28:C0:E6:47:47:61:87:92:3A:68:FB:49:A9:29:63:9C:64:89:22
        X509v3 Authority Key Identifier:
            F8:86:A0:6E:28:06:7B:2F:27:8D:4D:DB:BA:26:D2:DE:E4:7C:10:92
        Authority Information Access:
            CA Issuers - URI:http://127.0.0.1:8200/v1/pki/ca
        X509v3 CRL Distribution Points:
            Full Name:
              URI:http://127.0.0.1:8200/v1/pki/crl
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
    31:6e:aa:3b:ed:91:b9:37:c2:9f:8c:60:07:f9:64:c6:0c:a4:
    ad:96:b7:82:f3:d5:1a:f9:2f:af:fd:6d:33:e3:0a:b4:f5:04:
    61:c4:98:0d:8d:5d:42:2b:70:0e:17:9c:b2:d1:eb:95:dd:2b:
    b1:a4:c7:58:37:07:29:5a:1e:ea:c8:7e:a1:74:34:4f:b3:30:
    73:1d:c8:cd:b0:18:77:dd:f6:a0:72:aa:91:c3:4a:c6:28:12:
    4c:f5:50:1f:9f:54:5f:05:1c:39:33:ca:87:ee:25:51:9a:ae:
    21:f9:80:b0:6f:7d:7c:03:b2:64:58:b9:8c:e9:a0:16:37:fd:
    43:39:ed:d5:00:67:33:4c:e3:3a:7e:b1:b0:a0:0e:2a:98:24:
    58:b8:49:ab:61:dd:73:f8:f3:b8:61:70:0b:31:1c:f5:b2:08:
    0f:ac:df:11:60:e8:48:16:6d:9d:01:08:bd:8b:12:d8:7f:08:
    f5:58:ad:c2:d8:14:84:a2:b2:3e:c6:ac:32:40:fe:ef:87:20:
    00:70:99:8e:d8:e1:0c:b7:37:0d:60:80:52:81:16:87:4f:ce:
    51:8b:02:f7:0c:7b:3e:b3:08:bb:33:1f:75:06:fd:e5:59:eb:
    cc:9f:3f:b3:4d:61:f0:05:ba:29:f6:a1:a8:79:db:af:19:34:
    3c:66:c8:1a:d8:b5:2c:18:60:d4:2e:ae:c0:0e:cc:10:85:a7:
    21:d6:58:3b:96:ea:7c:5f:b1:29:43:48:f6:df:9c:a3:e8:8a:
```

```
        5d:4a:0a:a4:7f:9c:1f:6c:f4:1c:a9:84:cf:2c:d8:26:3d:d6:
        a4:98:b7:1c:65:0d:a1:e6:c0:7f:2d:67:76:0a:a5:35:05:0f:
        56:3c:fb:7b:20:02:b2:d4:29:0e:69:15:0a:ba:56:84:05:3a:
        fc:d1:a1:ab:21:7f:69:ce:2f:40:3d:ac:96:26:74:60:d7:9a:
        67:46:a6:d8:d1:d3:61:b3:1f:d3:9b:43:b3:4a:90:5c:bb:67:
        3a:32:10:67:3c:e0:98:ab:d4:cc:fa:b8:79:e1:58:f0:3e:91:
        2c:57:ac:13:45:2d:50:7a:3e:b9:a7:35:6d:04:39:73:54:dd:
        60:9e:f7:27:ca:f3:bb:b0:5c:ff:80:ed:c5:6e:48:6f:e4:13:
        5e:26:43:1f:72:77:26:d7:79:23:31:de:8b:ad:76:c9:8a:1b:
        5a:52:86:00:51:fe:54:62:84:34:4c:6d:21:a5:88:22:31:b0:
        ca:7d:ec:79:6a:cd:75:41:22:83:31:66:1d:3b:40:87:84:39:
        4d:44:cc:7b:bc:fc:72:c9:55:7c:59:7b:1a:3b:64:f4:79:2e:
        41:93:19:bb:42:e2:4a:75
```

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            26:ac:6d:f1:9a:f4:60:86:dc:0b:21:bc:53:aa:20:b4:7d:eb:99:31
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: CN = example.com
        Validity
            Not Before: Mar 18 16:58:04 2026 GMT
            Not After : Mar 15 16:58:33 2036 GMT
        Subject: CN = example.com
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (4096 bit)
                Modulus:
                    00:df:d1:d4:a6:de:be:15:04:33:4f:e8:96:04:fa:
                    db:8b:5f:9c:35:bb:83:1f:72:d4:0d:b7:5a:89:26:
                    59:ba:27:0d:ca:98:40:d9:b9:0c:33:62:84:b8:35:
                    c7:3b:35:34:2a:16:32:80:4c:82:be:b7:be:37:78:
                    c3:12:97:2f:90:08:3e:12:77:5f:bd:99:06:dd:81:
                    30:57:6f:c1:c7:fb:6f:5c:9a:b5:e3:96:09:f0:2d:
                    34:ec:16:03:37:54:df:7e:ee:bf:1c:6a:48:bc:35:
                    47:51:56:18:6e:f3:02:1e:28:7c:0a:a6:5f:10:1f:
                    00:74:89:92:51:de:06:36:eb:60:ff:bd:1a:61:d0:
                    8b:97:9e:6c:7c:b4:7f:3f:71:24:6b:5e:a7:cb:c6:
                    84:a2:86:36:d3:94:66:46:1c:4b:02:05:e8:c5:c6:
                    69:fe:2d:8f:4a:5f:2e:31:41:02:0e:da:66:f4:36:
                    2c:38:e5:f6:ce:8f:0b:a9:7b:49:0c:d1:24:dd:d8:
                    49:4c:01:e2:1a:f0:00:d0:39:4f:b6:b3:ef:e2:7d:
                    b5:95:c4:87:c5:ce:11:80:a6:9c:d7:6f:02:48:7d:
                    da:2c:12:16:73:6b:7c:c8:73:71:9b:2f:94:be:17:
                    f5:70:52:d0:77:26:c9:03:26:6c:16:d6:82:f0:3a:
                    59:b5:f3:e9:91:c9:14:3e:13:bc:d5:df:b1:e5:84:
                    9c:9a:ac:4f:d3:2b:ef:8b:fb:3d:57:9c:2d:64:2c:
                    00:a2:dd:b8:18:15:78:b5:c5:d9:71:46:4a:6f:b6:
```

```
            6d:5a:c9:76:6a:1b:7e:af:40:26:67:bc:a4:25:82:
            1a:5e:00:50:1e:1f:2c:e8:6e:03:bc:b4:cf:6e:f4:
            f1:f6:82:5b:3c:f6:c9:5f:6e:2b:67:b0:38:c1:df:
            d5:8d:74:5a:6a:9a:c4:ab:e5:4c:00:37:20:8d:4c:
            b4:ec:17:bd:13:6e:4b:75:7b:80:57:44:ee:82:73:
            9e:c3:0d:a7:25:3a:87:b6:b5:e8:12:3b:75:86:a2:
            5f:53:1a:43:cb:b2:f6:61:cb:9f:87:1d:a9:8f:4f:
            71:52:ae:9d:59:ba:17:94:24:e7:c7:dd:94:be:2e:
            f5:74:61:98:13:39:b3:2a:86:e8:95:97:d2:e3:18:
            f0:60:9a:2c:b7:fd:44:56:7b:17:c7:06:49:94:5b:
            b5:b1:60:8d:02:58:a6:2f:e9:32:ed:b4:8b:d4:fe:
            95:77:7e:97:dd:70:b0:e2:96:98:b6:13:ee:ae:ac:
            6b:e4:ef:64:bb:8b:2d:0b:25:b1:07:ae:4e:c9:9d:
            f2:ae:58:c2:05:0f:b0:39:a4:55:eb:25:17:c9:6b:
            68:59:9f
        Exponent: 65537 (0x10001)
    X509v3 extensions:
        X509v3 Key Usage: critical
            Certificate Sign, CRL Sign
        X509v3 Basic Constraints: critical
            CA:TRUE
        X509v3 Subject Key Identifier:
            F8:86:A0:6E:28:06:7B:2F:27:8D:4D:DB:BA:26:D2:DE:E4:7C:10:92
        X509v3 Authority Key Identifier:
            F8:86:A0:6E:28:06:7B:2F:27:8D:4D:DB:BA:26:D2:DE:E4:7C:10:92
        Authority Information Access:
            CA Issuers - URI:http://127.0.0.1:8200/v1/pki/ca
        X509v3 Subject Alternative Name:
            DNS:example.com
        X509v3 CRL Distribution Points:
            Full Name:
              URI:http://127.0.0.1:8200/v1/pki/crl
Signature Algorithm: sha256WithRSAEncryption
Signature Value:
    64:27:df:07:b5:29:17:4d:40:ab:04:0c:9c:02:df:a5:23:90:
    a1:98:66:02:79:87:1b:59:22:79:ff:80:72:05:3f:5d:eb:d0:
    9d:07:0e:3a:1f:b1:ff:f6:8b:18:e4:ca:b9:f6:67:1f:90:1a:
    fd:9a:d3:ea:e7:59:f6:1d:d3:0b:e0:6b:fe:37:82:8a:a5:6a:
    36:ed:4b:29:e0:24:3f:a8:62:80:fe:a4:68:46:37:4c:fc:d5:
    85:80:55:9f:d5:b4:21:c7:c2:6b:97:00:c1:37:be:b1:78:f7:
    07:d2:ff:a2:08:13:d9:8d:50:09:73:ca:e7:4d:17:77:90:f5:
    f8:3e:37:cd:98:89:1f:c6:12:80:6d:0e:4a:7a:66:0d:52:e1:
    7a:bd:b9:ef:8a:e7:65:d8:62:b4:19:7e:fd:8e:e3:28:6e:06:
    23:fa:a5:85:9e:fe:73:95:91:53:e2:28:0e:05:20:96:6d:e8:
    5d:cc:54:57:6f:4a:40:f6:0d:3b:72:a0:c2:56:49:f9:1a:dc:
    45:6b:c9:a3:64:6d:11:0d:9c:c3:65:63:fc:e0:0b:1e:ed:4a:
    2a:62:13:fa:20:13:36:a2:a6:e5:07:86:7f:dc:16:68:6d:7f:
    10:05:6c:27:cd:98:05:7d:99:41:66:2e:9c:e7:7c:62:cf:8d:
    f3:bb:a9:1e:12:4f:50:42:49:7b:4a:2c:79:75:af:0b:c0:27:
    b7:5a:e4:16:fe:af:c0:fa:14:bd:ae:5b:8d:1d:b3:5b:41:6f:
    38:3d:b1:2c:5a:3c:f6:0d:f5:24:06:ab:3d:8e:ca:0e:7d:4b:
    20:5a:a1:09:8d:48:b4:09:ee:22:d1:6f:dd:45:27:6c:54:5f:
    8b:a2:0b:71:05:68:cf:46:77:2c:12:66:15:54:26:fc:2f:74:
```

e9:1f:8c:01:af:10:4f:9d:c8:a8:26:af:4b:82:95:b5:18:83:
60:60:92:78:1d:fa:a3:b7:68:d5:16:17:c1:38:63:78:d6:c1:
eb:c9:4d:a6:f6:be:25:9e:45:a5:22:50:ec:35:3d:b1:f4:2b:
b7:ae:34:65:46:07:40:fb:07:5d:4f:ca:7e:cd:00:61:3b:f3:
e6:bc:17:81:4f:84:00:f5:21:64:61:4f:00:7b:7a:12:97:22:
57:de:4d:03:1c:f1:70:90:ed:1c:40:10:fa:67:e4:b0:fd:b6:
a3:3f:9b:90:5b:03:41:01:33:40:24:ad:e4:11:a0:50:05:5f:
ed:4a:ce:2f:3d:3d:44:b6:95:1f:54:c3:0d:79:8a:fa:33:be:
0d:4e:a9:fb:5c:b4:40:54:33:2f:4e:21:80:f2:50:c6:df:03:
c1:df:6d:7b:7f:f4:0e:65